SPLITTING A RING OVER THE OPEN PIECES OF A FINITELY GENERATED IDEAL BY ADDING ONE SURJECTIVE COMPONENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 31983-31990 (canonical tag 0C2A) and the counted proof is the proof environment at source lines 31992-32027, which immediately follows it and is its own proof.

Editorial note. The proof refers once, in its last paragraph, to the labelled lemma algebra-lemma-cover of another chapter for the glueing of functions. That label is not part of this chapter file, so the card carries one navigation stub whose pointer is the canonical tag list line of the snapshot; the stub only resolves the cross reference in the compiled document and carries no mathematics.

Editorial note. The lead-in sentence at source line 31981, calling the statement a simple glueing lemma, is outside the counted spans, as is the remark at source lines 32029-32030 pointing to another lemma for the construction of quotients by idempotents.

Editorial note. The chapter file pinned-original-source.tex carries only an input of the shared preamble on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus the native theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

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Allowed standard prerequisites (editorial).

- Localization at an idempotent splits a ring as a product of the localization at the idempotent and the quotient by the ideal generated by it. used at proof lines 32015-32018, where C'_{f_i} = A_{f_i}/(e_i) and the decomposition B = B_e times B/(e) for an idempotent e is applied. Identifies the localization of the constructed component C' at each generator with the complementary factor of C_{f_i}, which is what produces the product C times C'.
- An idempotent of a localization A_f can be written in the form a/f^n, and the representative can be adjusted so that it is itself an idempotent after multiplying by a power of f. used at proof lines 31997-32001, where e_i = a_i/f_i^n is written and the relation a_i^2 = f_i^n a_i is arranged.

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Produces the elements a_i in A whose ideal defines C' and makes the comparison of the two components at f_i computable.

- If two elements agree after inverting f_i f_j for all i and j then they agree after inverting the product of the generators, and an isomorphism that holds after localizing at f_1, ..., f_r holds after localizing at any element of the ideal they generate. used at proof lines 32002-32006 and 32020-32026. Gives the bounded relations (f_i f_j a_j)^N (f_j^n a_i - f_i^n a_j) = 0 and, at the end, transfers the isomorphism from the generators to every element of I.

Source reference. Original lemma 00EO: sources/stacks-chow/tags.txt, source lines 542–542. The final sentence uses the glueing of local isomorphisms of another chapter to pass from the generators f_1, ..., f_r to every element f of I; the whole construction of C', the exponent bookkeeping and the verification at each generator are inside the counted span, so the reference is a pointer to a standard glueing statement and not a substitute for the argument.

Lemma 0.2. *Let A be a ring and let I be a finitely generated ideal. Let $A \to C$ be a ring map such that for all $f \in I$ the ring map $A_f \to C_f$ is localization at an idempotent. Then there exists a surjection $A \to C'$ such that $A_f \to (C \times C')_f$ is an isomorphism for all $f \in I$.*

Proof. Choose generators $f_1, \ldots, f_r$ of I. Write

$$C_{f_i} = (A_{f_i})_{e_i}$$

for some idempotent $e_i \in A_{f_i}$. Write $e_i = a_i/f_i^n$ for some $a_i \in A$ and $n \geq 0$; we may use the same n for all $i = 1, \ldots, r$. After replacing a_i by $f_i^m a_i$ and n by $n+m$ for a suitable $m \gg 0$, we may assume $a_i^2 = f_i^n a_i$ for all i. Since e_i maps to 1 in $C_{f_i f_j} = (A_{f_i f_j})_{e_j} = A_{f_i f_j a_j}$ we see that

$$(f_i f_j a_j)^N (f_j^n a_i - f_i^n a_j) = 0$$

for some N (we can pick the same N for all pairs i, j). Using $a_j^2 = f_j^n a_j$ this gives

$$f_i^{N+n} f_j^{N+nN} a_j = f_i^N f_j^{N+n} a_i a_j^N$$

After increasing n to $n + N + nN$ and replacing a_i by $f_i^{N+nN} a_i$ we see that $f_i^n a_j$ is in the ideal of a_i for all pairs i, j. Let $C' = A/(a_1, \ldots, a_r)$. Then

$$C'_{f_i} = A_{f_i}/(a_i) = A_{f_i}/(e_i)$$

because a_j is in the ideal generated by a_i after inverting f_i. Since for an idempotent e of a ring B we have $B = B_e \times B/(e)$ we see that the conclusion of the lemma holds for f equal to one of $f_1, \ldots, f_r$. Using glueing of functions, in the form of Algebra, Lemma 00EO, we conclude that the result holds for all $f \in I$. Namely, for $f \in I$ the elements $f_1, \ldots, f_r$ generate the unit ideal in A_f so $A_f \to (C \times C')_f$ is an isomorphism if and only if this is the case after localizing at $f_1, \ldots, f_r$. $\square$